MPG No. 29.1.109 $m\bar{3}$ [Type I, cubic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{xx} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxyy} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxyy} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxyy} & Q_{xxxx} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_4^{(1)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_4^{(1)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_4^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & -Q_{xxyy} & 0 & 0 & 0 \\ -Q_{xxyy} & 0 & Q_{xxyy} & 0 & 0 & 0 \\ Q_{xxyy} & -Q_{xxyy} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}^{(3)}_{xyz}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{xxzz} & Q_{xxxx} & Q_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxzz} & Q_{xxxx} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}^{(1)}_0 + 2\mathbb{Q}^{(2)}_0 + 2\mathbb{Q}^{(1)}_4$$
$$Q_{xxyy} = 4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_0 - \mathbb{Q}^{(1)}_4$$
$$Q_{xxzz} = -4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_0 - \mathbb{Q}^{(1)}_4$$
$$Q_{yzyz} = \mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_0 - \mathbb{Q}^{(1)}_4$$
$$Q_{yzzy} = -\mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_0 - \mathbb{Q}^{(1)}_4$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_0$$
$$Q_{yzzy} = \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_0$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & 0 \\ 0 & Q_{xxxx} & 0 \\ 0 & 0 & Q_{xxxx} \\ 0 & 0 & Q_{yyzz} \\ Q_{yyzz} & 0 & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & 0 & Q_{yyzz} \\ Q_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}^{(1)}_0 + 2\mathbb{Q}^{(1)}_4$$
$$Q_{yyzz} = \mathbb{Q}^{(1)}_0 - \mathbb{Q}^{(1)}_4$$